

**The Hong Kong Polytechnic University**  
**Subject Description Form**

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| <b>Subject Code</b>                                   | AP30021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Subject Title</b>                                  | Optical Design with AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Level</b>                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Exclusion: AP30007                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Objectives</b>                                     | The objectives of this subject are to provide an up-to-date treatment of the optical design at an introductory level to students in the fields of applied optics, photonics, optical engineering and optoelectronics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Intended Learning Outcomes</b>                     | <p>On completing the subject, students will be able to</p> <ul style="list-style-type: none"> <li>(a) use the ray tracing method to solve the image formation problems; locate stops and pupils of an optical system consisting of lenses, mirrors, and prisms;</li> <li>(b) explain the formation of the first-order aberration and the third-order aberration; identify and calculate various types of monochromatic aberrations and chromatic aberrations;</li> <li>(c) solve complex optimization problems in optical systems with the assistance of AI technologies; and</li> <li>(d) analyze and propose solutions to practical problems related to optics and light illumination.</li> </ul>                                        |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <p>paraxial optics, ray tracing, lens system, physical optics, optical materials.</p> <p>first-order aberrations, third-order aberrations, chromatic aberrations, image quality measures, optical transfer function (OTF), OTF computation.</p> <p>merit function, optical design procedures, lens design optimization, tolerance analysis.</p> <p>radiation, color and color rendering, practical light sources and laws of illumination, interior and exterior illumination designs, methods for calculating illumination.</p> <p>mini-projects on the design and optimization of singlet/doublet, spectrometer, beam expander, double Gauss camera lenses, Newtonian telescope, Maksutov telescope, zoom lens, and chromatic prism.</p> |
| <b>Teaching/Learning Methodology</b>                  | <p><b>Lecture:</b> The course contents will be delivered through lectures. Active participation in discussion by students will be encouraged. Tests and quizzes will be given to class at appropriate intervals to consolidate students' understanding of the acquired knowledge as well as to enhance their problem-solving skills.</p> <p><b>Practical assignment:</b> Students will be given opportunities for hands-on operations on advanced optical design software. They are expected to gain direct experience in making independent optical design. Through the process students can enhance their analytical power, problem solving skills, conceptual understanding, critical thinking and desire for life-long learning.</p>   |

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| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b> | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                         | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |       |
|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             | a                                                                              | b | c | d     |
|                                                                        | (1) Continuous assessment                                                                                                                                                                                                                                                                                                                                                                                                                                 | 60          | ✓                                                                              | ✓ | ✓ | ✓     |
|                                                                        | (2) Practical Test                                                                                                                                                                                                                                                                                                                                                                                                                                        | 40          | ✓                                                                              | ✓ | ✓ | ✓     |
|                                                                        | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100         |                                                                                |   |   |       |
|                                                                        | <p>Continuous assessment including assignments, laboratory exercises, quizzes and test(s). The continuous assessment is designed to monitor the study progress of the student. They also serve to provide student with a mechanism of self-evaluation of learning achievement. The Practical Test will be used to assess the knowledge acquired by the students; as well as to determine the level of attainment of the prescribed learning outcomes.</p> |             |                                                                                |   |   |       |
| <b>Student Study Effort Expected</b>                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |   |       |
|                                                                        | • Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                |   |   | 24 h  |
|                                                                        | • Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                |   |   | 21 h  |
|                                                                        | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |   |   |       |
|                                                                        | • Self-study                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                |   |   | 74 h  |
|                                                                        | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                                |   |   | 120 h |
| <b>Reading List and References</b>                                     | Robert E. Fischer, “Optical system design”, 2 <sup>nd</sup> Edition, McGraw-Hill 2008.                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                |   |   |       |
|                                                                        | Eugene Hecht, “Optics”, 5 <sup>th</sup> Edition, Pearson Addison Wesley, 2016                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                |   |   |       |
|                                                                        | Herbert Gross, “Handbook of Optical Systems”, WILEY-VCH Verlag GmbH & Co. KGaA, 2005                                                                                                                                                                                                                                                                                                                                                                      |             |                                                                                |   |   |       |
|                                                                        | Optical design software used: Zemax OpticStudio                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |   |   |       |